Homework 6
Due: Wednesday, 10th July 2024 at 1:00 PM

Instructions. Provide your answers **and an explanation** to the following questions and *submit as a single PDF on Gradescope* by the due date and time listed above. *Assign pages accordingly when you submit. It is recommended that you type as much as you can to ensure that your solution is legible. Illegible solutions will not receive credit.*

Note on Academic Dishonesty. Although you are allowed to discuss these problems with other people, your work must be entirely your own. It is considered academic dishonesty to read anyone else's solution to any of these problems or to share your solution with another student or to look for solutions to these questions online. See the syllabus for more information on academic dishonesty. If it is determined that you have violated any of these guidelines, you will be reported to the Dean of Students and the recommended sanction will be a failing grade in the course.

Grading. Some of these questions may be graded for accuracy and some for completion. Each question is worth 10 points.

Question 1.

Show that the set S defined by $1 \in S$ and $s + t \in S$ whenever $s \in S$ and $t \in S$ is the set of positive integers.

Question 2.

Consider the recursively defined function with two parameters m and n. The basis step is $a_{1,1}=5$. The recursive step is:

$$a_{m,n} = \begin{cases} a_{m-1,n}+2 \text{ if } n=1 \text{ and } m>1 \\ a_{m,n-1}+2 \text{ if } n>1 \end{cases}$$

Show that $a_{m,n}=2(m+n)+1$ when m and n are positive integers.

Consider an inductive definition of a version of **Ackermann's function**. This function was named after Wilhelm Ackermann, a German mathematician who was a student of the great mathematician David Hilbert. Ackermann's function plays an important role in the theory of recursive functions and in the study of the complexity of certain algorithms involving set unions. (There are several different variants of this function. All are called Ackermann's functions and have similar properties even though their values do not always agree.) Please use the following variant of Ackermann's function in this homework:

$$A(m, n) = \begin{cases} 2n & \text{if } m = 0 \\ 0 & \text{if } m \geq 1 \text{ and } n = 0 \\ 2 & \text{if } m \geq 1 \text{ and } n = 1 \\ A(m-1, A(m, n-1)) & \text{if } m \geq 1 \text{ and } n \geq 2 \end{cases}$$

Questions 3–4 involve this version of Ackermann's function.

Question 3.

Find these values of Ackermann's function.

a) A(1, 0) b) A(0, 1)

c) A(1, 1) d) A(2, 2)

Question 4.

Show that $A(1, n) = 2^n$ whenever $n \geq 1$.

Question 5.

Compute the GCD of 20 and 52 using the brute force method that we discussed in class. Provide all divisors of 20 and 52. Provide the list of common divisors. And finally provide the GCD.

Question 6.
Compute the GCD of 98 and 154 using the brute force method that we discussed in class. Provide all divisors of 98 and 154. Provide the list of common divisors. And finally provide the GCD.

Question 7.
Compute the GCD of 20 and 52 using the Euclidean algorithm. Show each step of the algorithm.

Question 8.

Compute the GCD of 98 and 154 using the Euclidean algorithm. Show each step of the algorithm.

Question 9.

Give a recursive algorithm for finding a **mode** of a list of integers. (A **mode** is an element in the list that occurs at least as often as every other element.) You can assume that the list contains at least one integer.

Questions 10 to 12 are about mergesort. For each of the questions, please show how the problems get divided into two parts and how they got merged. For an odd number of elements you can put one more element either in the right part or in the left part but be consistent.

Question 10.
Consider the list 1, 5, 2, 3, 6, 4. Execute mergesort on this list and show each step of the algorithm.

Question 11.

Consider the list 3, 5, 7, 3, 6, 7, 1, 4. Execute mergesort on this list and show each step of the algorithm.

Question 12.

Consider the list 8, 17, 6, 2, 16, 5, 13, 8, 11. Execute mergesort on this list and show each step of the algorithm.

Question 13.

Consider the following program:

procedure *multyply_neg_one*(n):

```
    m := -1
    i := 1
    while i<n:
        i := i + 1
        m := m.(-1)
```

Show that after executing the program for a particular value of $n>0$, $m = (-1)^n$

Question 14.

Consider the following program:

```
procedure sum(n):
    s := 1
    i := 1
    while i<n:
        i := i + 1
        s := s + i
```

Show that after executing the program for a particular value of $n>0$, $s = n(n+1)/2$

Question 15.

Consider the following program:

```
procedure iterative_fibonacci(n)
    x := 0
    y := 1
    i := 1
    while i < n
        i := i + 1
        z := x + y
        x := y
        y := z
```

Show that after executing the program for a particular value of $n>0$, $y = f_n$, the n-th element of Fibonacci sequence.